Answer key: Lesson 6, Tens Jump, Ones Hop

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. These are Mei's sticks. She gets another bundle of 10 sticks. Fill in the blanks.

Answer: 3 bundles, 7 singles, 37 sticks

The picture shows 2 bundles and 7 singles, which is 27. One more bundle makes 3 bundles, and the 7 singles stay. 30 and 7 is 37.

Watch for: Adding the bundle to the singles (28).; Forgetting the old bundles (17).

S2. In the box, draw 3 tens as 3 tall bars and 4 single sticks. Next, draw 2 more tall bars. How many sticks are there altogether?

Answer: 54 (5 bars drawn and 4 singles)

3 bars and 4 singles is 34. Two more bars add 20, so there are 5 bars and 4 singles: 54.

Watch for: Adding the 2 to the singles (36).; Drawing 2 sticks instead of 2 bars of ten.

S3. Jump forward. Fill in the blanks.

Answer: 33, 63, 81, 67

(a) 2 tens and 1 ten make 3 tens: 33. (b) 2 tens and 4 tens make 6 tens: 63. (c) 5 tens and 3 tens make 8 tens: 81. (d) 7 is 7 ones; add 6 tens to get 67.

Watch for: Changing the ones digit (23 + 40 = 27).; (d) writing 13 (7 + 6).

S4. Jump or hop. Fill in the blanks.

Answer: 24, 38, 55, 69

(a) 3 tens take away 1 ten is 2 tens: 24. (b) 5 tens take away 2 tens is 3 tens: 38. (c) 3 is a hop: 8 ones take away 3 is 5 ones: 55. (d) 7 is a hop forward: 2 and 7 make 9 ones: 69.

Watch for: Taking the 3 from the tens digit (58 - 3 = 28).; Hopping the wrong way in (d).

S5. Ali has 68 pencils. He gives 3 boxes of 10 pencils to his friend. How many pencils does Ali have left?

Answer: 38 pencils

3 boxes of ten is 30 pencils. 68 - 30 = 38: jump back 3 tens (58, 48, 38). The 8 ones stay.

Watch for: Taking away 3 (65).; Adding instead of taking away (98).

S6. Ben says 45 + 20 = 47. Fill in a to c.

Answer: No. The right answer is 65. He added 2 ones instead of 2 tens.

20 is 2 tens, so Ben should jump two tens: 45, 55, 65. He added 2 ones and got 47. A jump changes the tens digit.

Watch for: Saying he is correct because 20 has a 2 in it.; Not saying what he mixed up (any clear sentence about tens and ones is fine).

S7. Start at 36. On a number line in the box, draw 2 jumps of ten. Where do you land?

Answer: 56 (two jumps of ten from 36: 46, then 56)

36 + 10 = 46, then 46 + 10 = 56. Two jumps of ten land on 56. The 6 ones never change.

Watch for: Drawing 2 small hops (38).; Changing the ones digit.

Think harder (page 3)

T1. Add 34 + 25. Fill in the blanks.

Answer: 59 (jump 20 to 54, then hop 5 to 59)

25 is 2 tens and 5 ones. 34 + 20 = 54. Then 54 + 5 = 59. Check: 3 tens and 2 tens are 5 tens; 4 ones and 5 ones are 9 ones.

Watch for: Adding only the tens and stopping at 54.; Adding 2 + 5 as 7 tens (74).

T2. Subtract 68 - 24. Fill in the blanks.

Answer: 44 (jump back 20 to 48, then hop back 4 to 44)

24 is 2 tens and 4 ones. 68 - 20 = 48. Then 48 - 4 = 44.

Watch for: Taking 24 away from the ones only.; Hopping forward instead of back.

T3. Both frogs start on 12. Frog A jumps 3 tens, then hops 4. Frog B hops 4, then jumps 3 tens. Fill in a to c.

Answer: Yes, both land on 46 (30 and 4 are added in either order)

Frog A: 12 + 30 = 42, then 42 + 4 = 46. Frog B: 12 + 4 = 16, then 16 + 30 = 46. They land on the same number: adding the same jumps and hops in a different order gives the same total.

Watch for: Saying Frog B lands on 16 because the hops came first.; Adding 3 + 4 and landing on 19.

T4. Ali starts on a number. He jumps forward 3 tens and ends on 78. Where did Ali start?

Answer: 48 (jump back 3 tens from 78)

Undo the 3 jumps of ten: 78, 68, 58, 48. Check: 48 + 30 = 78.

Watch for: Adding 30 to 78 (108).; Taking away only 3 (75).

Go further (page 4)

G1. Mei is on 7 and must get to 97. Each time she can jump 10 or jump 20. Many answers are right. Fill in a and b.

Answer: Any two different routes of 10s and 20s from 7 to 97, e.g. 7, 17, 27, 37, 47, 57, 67, 77, 87, 97 (nine jumps of 10) and 7, 27, 47, 67, 87, 97 (four of 20 and one of 10).

Every jump adds 10 or 20, and all the jumps together add 90 (97 - 7). Examples: nine jumps of 10; or 20, 20, 20, 20, 10 landing on 27, 47, 67, 87, 97. The ones digit stays 7 on every landing.

Watch for: Landing on a number that is not 10 or 20 more than the one before.; Two routes that are the same jumps in the same order.; Stopping at 87 or going past 97.

G2. Now Mei wants the fewest jumps from 7 to 97. She can jump 10 or jump 20. Fill in a and b.

Answer: 5 jumps: four jumps of 20 and one jump of 10 (any order). Four jumps travel at most 80, so she reaches only 87.

She must travel 9 tens. Four jumps of 20 make 8 tens. One jump of 10 makes the last ten. That is 5 jumps. Four jumps can travel at most 80, which only reaches 87, so 5 is the fewest.

Watch for: Saying 4 (80 only reaches 87).; Saying 9 (only using jumps of 10).

G3. Start at 25. Jump forward 3 tens, jump back 1 ten, then jump forward 2 tens. Then hop and land on 69. Fill in a to c.

Answer: 55, 45, 65, then a hop of 4 to 69. The three jumps are the same as 4 tens forward.

25 + 30 = 55. 55 - 10 = 45. 45 + 20 = 65. To land on 69 the hop is 4. Short way: 3 - 1 + 2 = 4 tens forward, so 25 + 40 = 65.

Watch for: Jumping forward on the jump back.; Saying the hop is 44 (69 - 25).; Saying 6 tens for (c) (adding 3 + 1 + 2).

Investigation (page 5): Jump road

From 6 you land on 16, 26, 36, 46, 56 and so on: every number ends in 6, so you never land on 50. Jumps of ten never change the ones digit. Only a start number that ends in 0 (0, 10, 20, 30, 40 or 50 itself) can land on 50.

Look for: A child who tests more than one start number, says the ones digit stays the same, and gives a reason using tens and ones (a jump is whole bars). A child who says start 20 can land on 50 because 20 + 30 = 50.

Exit ticket (page 5)

E1. Add.

Answer: 76

4 tens and 3 tens make 7 tens. The 6 ones stay: 76.

Watch for: Writing 49 (adding 3 to the ones).; Writing 16.

E2. Subtract.

Answer: 44 and 71

(a) 30 is 3 tens: jump back to 44. (b) 3 is a hop: hop back to 71.

Watch for: Doing the same for both parts.; (b) writing 44.

E3. Mei is on 52. She jumps forward 2 tens, then hops forward 3. Fill in the blanks.

Answer: 72 and 75

2 tens forward from 52 is 72. Then a hop of 3 forward is 75.

Watch for: Writing 54 (adding 2 to the ones).; Stopping at 72.

Sneak-peek puzzles (not in the workbook)

1. 36
2. No. 43 + 20 = 63. 20 is 2 tens, so the tens digit changes, not the ones.